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Machine Learning-Based Feature Selection Intrusion Detection System for Enhancing IoT Network Security

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Abstract

The rapid proliferation of Internet of Things (IoT) devices has introduced unprecedented connectivity but simultaneously amplified cyber-security risks. The IoT networks are vulnerable to breaches and cyber attacks because of compromised security protocols and inadequately secured IoT devices. These devices have constrain over security update and lack of encryption make them prone to cyber attacks. This paper presents a comprehensive study on developing a robust intrusion detection system for IoT networks by leveraging an ensemble machine learning approach. The methodology encompasses a complete pipeline from raw traffic capture and sophisticated feature extraction using CICFlowMeter, through meticulous data pre-processing to advanced feature selection and the implementation of diverse machine learning models. Further it presents a strategic machine learning-based IDS framework using a stack classifier incorporating optimized feature selection to enhance detection accuracy and efficiency. Modelling using the K-Best algorithm to select features and extract the top 15 most important ones and building an ensemble model are the two mainstays of this method. Classification algorithms including Random Forest (RF), Naive Bayes (NB), Support Vector Machine (SVM), K-Nearest Neighbors (KNN) are compared with proposed custom-built Stack Classifier by considering the ‘Ton IoT dataset’. The findings unequivocally demonstrate the superior performance of the Stack Classifier, achieving significantly higher accuracy, precision, recall, and F1-score, thereby identifying it as the optimal model for enhanced intrusion detection.

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1. Introduction

A new computer paradigm has begun with the launch of the IoT and is changing our lives. There are many different uses for IoT devices. That includes transportation, smart homes, healthcare, industrial production, security, supply chain management, blockchain monitoring [1]. In 2021, according to the study [2] [3] there were 13.8 billion IoT devices in use, and more than 30.9 billion units of IoT networks will reportedly be in use by 2025. This sector is expanding at a rate of more than 55% annually due to its fast expansion, reliance on IoT devices, and commercial possibilities [3]. IoT gadgets are in high demand owing to the many useful functions they offer. Possible investors are interested in this demand. By developing and selling new uses for IoT devices, entrepreneurs are increasing the appeal of the field and making our lives easier and better [4]. However, it has opened the door for cybercriminals to take advantage of the loop-holes in IoT security. Cybercriminals have found the Internet of Things a lucrative target because of its potential for rapid company expansion. The exponential increase in cyberattacks against IoT devices can be attributed to this [5].

IoT device production that pays little or no attention to data protection. There are many IoT devices which are vulnerable to cyberattacks even though they are completely functional in various services [6]. There is a wider threat landscape surrounding IoT devices. In most cases, other appliances and systems can be connected to IoT devices. Medical Internet of Things (MIoT) is one of the fields within the Internet of Things that is expanding at a rapid bound. MIoTs are delicate due to the direct impact they have on patients' health [7]. The security loop - holes in the Internet of Things poses a serious threat. Attacks and vulnerabilities in IoT devices can be exploited in several ways [8]. Unless we include cyber physical attacks, all cyberattacks involve network connectivity. IoT devices are systems with computational constraints. That is why it is not feasible to secure the insecure IoT environments using typical cyber-security procedures [9]. Furthermore, traditional rule-based security measures are inadequate for the IoT context due to the wide variety of potential vulnerabilities [10]. It is not easy to identify unusual data from various data streams. Furthermore, these situations call for ML-based solutions. Deep neural networks excel at sorting data according to specific attributes, analyzing massive amounts of data, and finding patterns and relationships among them [11].

To strengthen the safety of IoT devices, machine learning-based intrusion detection methods are needed. When it comes to monitoring and analyzing data, they are tops. They can handle massive amounts of IoT data with ease. Algorithmic training allows them to spot suspicious trends, such as unexpected sensor readings, attempts at unauthorized access, or network activity, and alert the appropriate authorities [12]. Rapid security responses and proactive threat avoidance are made possible with continuous monitoring, which identifies risks in real-time. In the current state of IoT security, this strategy is essential for successfully countering the dynamic nature of cyberattacks. In addition, intrusion detection systems that use behavior or anomaly detection do more than just find certain attack signatures. Rather, their main objective is to identify and analyze patterns of behavior that are harmful or unconventional. These state-of-the-art systems analyze large data sets and network traffic using statistical analysis, AI, and machine learning techniques; their main objective is to identify any anomalies [13].

Security breaches are more likely to occur when IoT devices share networks with other business equipment. Network segmentation is suggested as a means of reducing the impact of this danger. Limiting the attack surface and isolating IoT devices from other devices helps reduce the impact of security breaches. Separation of the IoT devices from the rest of the network is necessary to achieve this goal. As shown in the study in [14], machine learning algorithms can be used to keep an eye on data transfers between IoT devices and other devices on the network, allowing for the rapid detection of any intrusion attempts [15]. Machine learning-based predictive analysis may look at past data and spot patterns that could lead to security breaches, even insider threats that are difficult for standard approaches to detect [16]. While traditional intrusion detection systems (IDS) have limitations in the world of big data, machine learning has allowed for more effective threat detection and classification, which has allowed for better monitoring of new threats. To make matters worse, processing massive amounts of data requires sophisticated technology and algorithms powered by AI [17].

To combat cybersecurity risks and ensure safety, AI is an encouraging approach. The IoT networks, including servers, clouds, sensors, devices, and applications running in the setup as shown in Fig. 1. can be adequately protected by deploying an anomaly-based IDS [18].

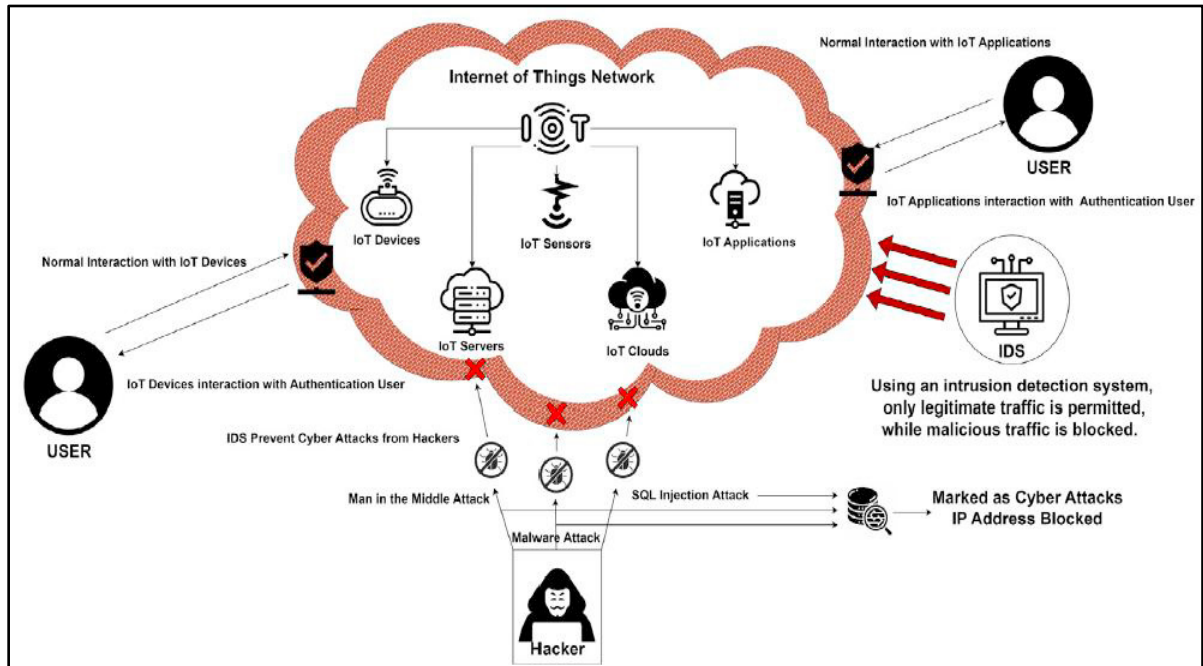


Fig. 1. Protecting the IoT network with IDS

The drive of this examination is to grow a state-of-the-art stack classifier model based on heterogeneous machine learning to improve intrusion detection approaches. Refining the general presentation and accuracy of intrusion detection systems is highly prioritized, especially when it comes to finding and stopping security breaches. In this research, Investigation to compare and contrast a number of different intrusion detection algorithms and methodologies that use machine learning is executed.

Here is how the rest of the paper is organized: In Section II, background information and a comprehensive review of relevant studies are included. The approach and its implementation are extensively covered in Section III. Section IV describes the results and discussions of the work. Section V outlines a summary of the paper and some avenues for further research.

2. Related Work

The implementation of IoT devices has simplified our daily survives. It has completely transformed many industries. Despite the impressive rate of development of IoT devices, concerns about cybersecurity have become a major obstacle to the rapid adoption of this technology [19]. When an attacker compromises an Internet of Things device, they get access to not only the device itself but also any other services it offers. Applications that make advantage of IoT devices are those that are closer to people's daily lives and the things they do for a living. Because of this, it has become as a popular target for cybercriminals [20]. In addition, several production facilities use IoT devices as their control unit [21]. If there is a security breach, an attacker can access the entire control system. Such attacks can even affect connected smart phones [22]. More and more cybercriminals are targeting IoT environments because of these appealing aspects of IoT networks. To reduce the frequency and severity of assaults on IoT systems, it is crucial to create an intrusion detection system based on deep learning [23].

Intrusion Detection Systems (IDS) serve as foundational components for safeguarding contemporary Information and Communication Technology (ICT) systems, particularly within the burgeoning IoT landscape. [24]. The network operation was described using Shapley Additive Explanations (SHAP). Understanding the reasoning behind their deep learning-based IDS decision is a primary goal of this approach [25]. With an F1 score of 98.03%, this framework shows an accuracy of 99.55%. However, the IoV, a subset of the IoT, is the main focus. According to [26], fog and edge computing environments are ideal for intrusion detection systems relying on machine learning. Regardless of the

number of devices connected to an IoT network, the suggested intrusion detection system could be a viable solution. At the same time, the 99.15% accuracy achieved [27] makes one wonder if overfitting is a problem that has not been tackled using alternative datasets. This method cannot compare to the suggested intrusion detection system in terms of reliability and robustness [28].

Designed to safeguard industrial IoT devices, an DL-based framework called DeepIIoT achieves 99% accuracy on the test data set. They focus on industrial-IoT, particularly in relation to water treatment plants, nuclear power plants, and electricity distribution networks. The WUSTL-IIOT-2021 dataset was used for both training and experiments [29]. Like its predecessor, the suggested IDS make use of DL. Nevertheless, it does not depend on any particular network due to its novel intermediate communication structure. Industrial IoT is only one area where the proposed IDS could be useful. Additionally, as a rule-based detection system, DeepIIoT is based on signatures. The difficulty of regularly updating the signature database is a major drawback. There is no such restriction in the proposed system [30]. A smart network-IDS using a feature fusion approach based on recurrent deep learning and an ensemble metaclassifier is proposed in [31]. According to [32], this method is 99% accurate. The suggested system brings a novel contribution to the field of IDS due to its simple network design, although it is more accurate than the suggested method. A new approach to intrusion detection for communication networks has been developed, which relies on Artificial Neural Networks (ANN) and is based on the mining of association rules [33]. Even the suggested method relies on ANN. But its network architecture is custom-made to detect intrusions in IoT networks.

The potential uses of DL strategies for IoT security have been covered in [34]. [35] highlighted the use of DL for IoT privacy. On the other hand, it is easier to make intrusion detection systems that are network-specific and based on rules. Within the specified network, they function admirably. Compared to [36], a comparable method called INTI performs 7% worse than [36]. It is very important to have the right intrusion detection system (IDS) in different setups in their study on active analysis of changes in rule-based systems [37]. The quantitative analysis of the paper supports the central premise of the methodology, that is, the need for an IDS for IoT networks that is independent of network protocols. IoT security have found new ways to improve intrusion detection systems. with an importance on improving security within the IoT network framework [38]. To combat intrusions and demonstrate enhanced confidentiality and authenticity, they provide an efficient pairing-free identity-based system that makes use of elliptic curve cryptography [39]. The effectiveness of the scheme in strengthening security within the cloud-assisted IoT environment has been confirmed by performance analysis and comparisons. Furthermore, several techniques were used to improve the performance of the intrusion detection system (IDS) in [40], with a primary emphasis on analyzing large amounts of traffic data from the network.

With ensemble learning approaches such as gradient boosting and random forest, the [41] investigated the detection of intrusions into the IoT. Improving detection precision while reducing false positives was the focus of their research. The authors of [42] presented an anomaly-based plan for IoT networks in their study. They use the RF algorithm to distinguish between typical and unusual traffic. The authors of the study in [43] created a model for classifying the devices of the Internet of Things (IoT) used in smart homes and achieved an impressive 99.79% accuracy. The model dynamically classifies devices based on 13 characteristics of network traffic, using logistic regression and supervised machine learning; this provides a solid foundation for monitoring a variety of IoT settings [44]. Typical network activity and security threat instances are examples of tagged data used to train supervised learning algorithms. The algorithm is able to identify comparable dangers in real time after training. A popular supervised learning method for intrusion detection is SVMs [45]. SM accomplishes its goal by distinguishing between typical and unusual network behavior. It sounds an alert if it detects any network traffic that does not fit this pattern.

2.1 ToN-IoT Dataset

The telemetry data of IoT devices, operating systems and network data all together represented by ToN-IoT dataset which is new generation dataset represents industrial IoT and IoT 4.0 environment. The key purpose behind origin of this dataset is to evaluate the various artificial intelligence and cyber security application-based approaches and algorithms. The Cyber Range and IoT Labs of UNSW Canberra at the Australian Defence Force Academy (ADFA) Developed this dataset to test the complexity and scalability characteristics of industrial IoT and Industry 4.0 which is expected to a foundation for cyber security research. This dataset is freely available for academic research.

ToN-IoT dataset includes the data captured from various IIoT sensors at device level like weather sensors, Modbus sensors, smart refrigerators, garage door sensors, GPS trackers, motion lights, and thermostats. At operating system level, the activity record of disk usage, process execution, processor load, memory consumption, and network activities are collected and network data including Packet Capture files, log files, and CSV files is provided at network level. Thus, this ToN-IoT dataset is rich in cyber-attack events like Reconnaissance and scanning, Malware and botnet attacks, Network disruption attacks and authentication-related attacks.

3. Proposed Methodology

The design of an efficient Internet of Things intrusion detection system often follows a methodological pipeline that is structured, as shown in Fig. 2. This procedure includes a number of essential stages, beginning with the initial data gathering and feature engineering, continuing with the painstaking data pre-processing and balancing, and concluding with the training, evaluation, and optimization of the system.

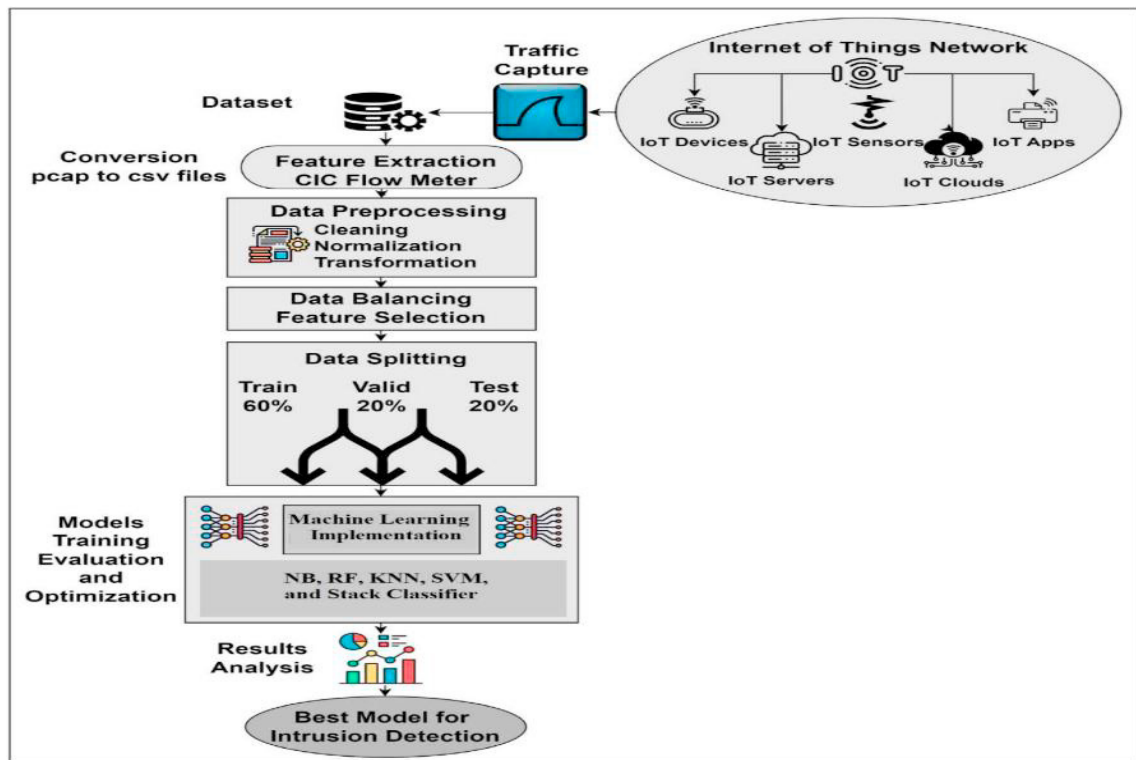


Fig. 2. Work flow of the proposed models

The network traffic data collected from IoT devices, sensors, servers and cloud infrastructure. This information is often voluminous and complex but rich in information so using tcpdump tool the raw network packets are stored in PCAP type files which afterward converted into manageable and analyzable CSV format files. Then in second stage the feature extraction is performed using CIC Flow Meter which generates the bidirectional network flow from PCAP files and identifies the to and fro directions of network communication. It creates a structured CSV file which includes the columns for flow IDs, source and destination IP addresses, port numbers, protocols, and the extensive array of extracted network traffic analysis features. This CICFlowMeter has several limitations such as conversion data loss, incorrect flow descriptions, incoherent timestamps and duplicate data entries. To overcome this, data pre-processing and balancing plays a crucial role and ensure the data uniformity. In pre-processing, missing, redundant and irrelevant values are cleaned then features are first normalized on similar scale and then transformed to modify the distribution and to stabilize variance. In the later stage the data is analyzed for imbalance and feature selection. Feature selection is a critical step in pre-processing as it helps in reducing the dimensionality by selecting most relevant features in the dataset.

3.1. Implementation

In the implementation phase, four supervised ML classifiers: Nave-Bayes (NB), Random-Forest (RF), SVM, K-nearest neighbor (KNN) and an ensemble method known as stack classifier are considered. Each considered algorithm is trained on the training data and validated through iterations for fine tuning. This iterative process of training, evaluation, and optimization is crucial for achieving high detection accuracy and minimizing errors. The iterative nature of model training, evaluation, and optimization reflects the empirical and adaptive nature of developing effective IDS.

For the security of IoT, this study incorporates ML intrusion detection algorithms. To evaluate the TON-IoT data set, a stack classifier is used in conjunction with four supervised ML classifiers: Nave-Bayes (NB), Random-Forest (RF), K-nearest neighbor (KNN), and SVM. After painstakingly collecting data from the TON-IoT dataset, the research procedure proceeds to prepare the data so that classifiers can use it [46]. The data set is divided into two parts: the train set and the test set. The stack classifier, NB, RF, KNN, and SVM are all trained in the training set [47]. Fig. 2 shows the process that identifies the best classifier for further investigation.

Their proven effectiveness in managing various patterns and complexities within IoT data is the driving force behind the adoption of traditional ML strategies, such as NB, RF, KNN, and SVM. The ensemble model benefits from the addition of these models, which are well known for their resilience and ability to discern complex correlations in the data [48]. A comprehensive strategy is necessary to tackle the varied types of attacks against IoT networks. These classic models are now part of the ensemble, guaranteeing that all possible infiltration situations are covered. Improving intrusion detection with an ensemble approach is shown in Fig. 3., which also shows how the ML models used and ensemble learning work together. Several criteria were considered during the selection process to ensure the usability and representativeness of the TON-IoT dataset. This dataset contains samples of both typical and atypical traffic, the latter coming from a variety of places such as hostile assaults, broken devices, and odd user actions.

$$\text{Let the data set } D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \quad (1)$$

Where, $x_n \in R_m$ and $y_n \in \{0, 1\}$

- *Feature selection:*

$$\text{Information Gain (IG): } IG(A) = H(Y) - H(Y|A) \quad (2)$$

Where, $H(Y)$ is entropy of labels and $H(Y|A)$ is the conditional entropy given attribute A .

- *Recursive Feature Elimination (RFE):*
$$= S_k \frac{1}{T} \sum_{t=1}^T \text{Accuracy}(M_t, F_k) \quad (3)$$

Where, M_t is the model at fold t and F_k is subset of K features. The optimal feature set F^* is

$$F^* = \arg \max_{F_k} S_k \quad (4)$$

In model training and evaluation, RF, NB, SVM, KNN and stack classifier are used. Each model trained on F^* and evaluated using following standard metrics.

TP - True Positive, TN - True Negative, FP - False Positive, FN - False Negative

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

$$F1 - \text{Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

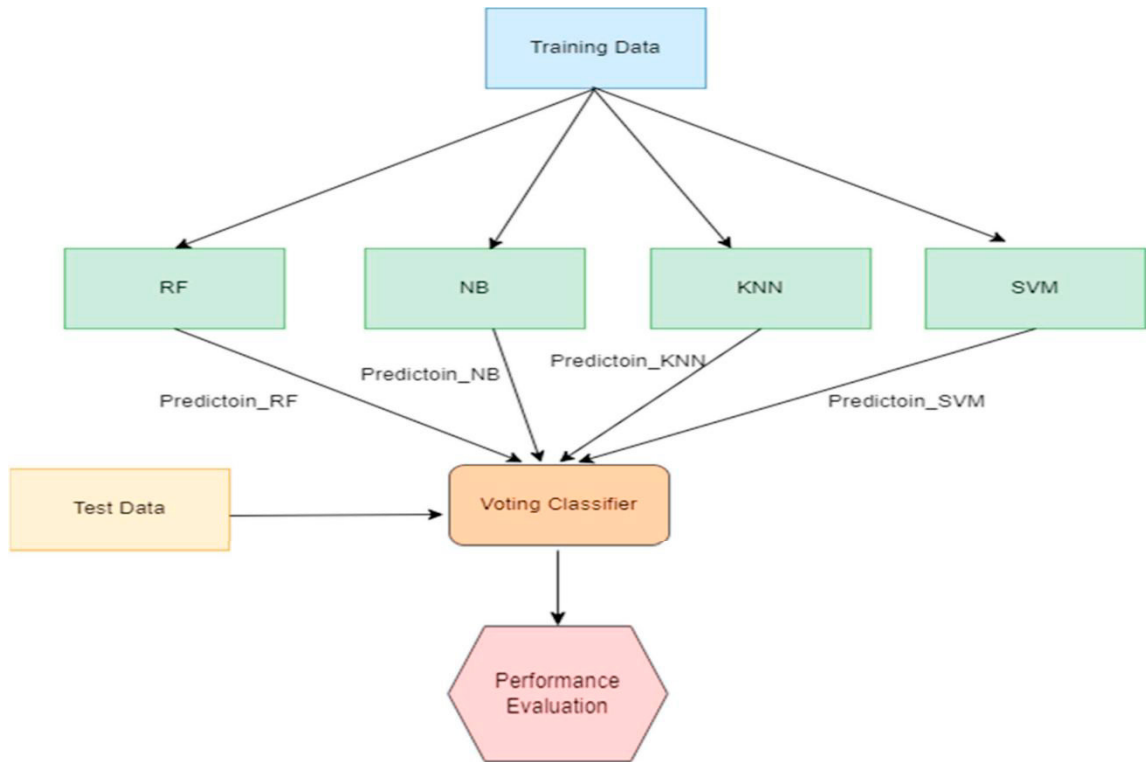


Fig. 3. Integrating ML Models and Ensemble Learning for Enhanced Intrusion Detection Implementation

3.2. K-Best feature selection technique

Optimizing ML model's performance is greatly assisted by the K-Best feature selection technique, which systematically identifies the top k most relevant characteristics from the provided dataset. This approach deliberately decreases the dimensionality of the data, improving its interpretability and reducing the likelihood of overfitting [49]. The scoring function, an application of the ANOVA F-value between label/outputs and features/inputs, is used to objectively evaluate features in the Select K Best technique. The 15 most important characteristics are chosen by instantiating a Select K Best object with the scoring function. After that, the fit transform method is used to make sure the feature selection is properly adjusted to the data, keeping only the top features according to their ANOVA F-value scores. The model's predicted accuracy is greatly enhanced by this painstaking procedure.

The top 15 features were elected using the K-Best process based on their significant impact on IoT security. Because of their remarkable ability to differentiate between benign and harmful activity within IoT networks, these characteristics are deemed increasingly crucial [50]. The features that are highly relevant to intrusion detection are given precedence by the criterion. One way to strengthen defenses against ever changing threats is by using the K-Best algorithm, which is a precise method of identifying these important qualities [51]. This open and careful feature selection procedure strengthens our security framework by ensuring that the model only considers the most important features of IoT data. Integrating the K-Best feature selection technique becomes crucial to optimize models. Important for intrusion detection, in particular, this thorough review ensures that the model zeros in on the most pertinent characteristics of IoT data [52]. Ensemble modeling, when seamlessly integrated, uses the combined wisdom of multiple machine learning models to provide strong protection against a wide range of incursion situations. An advanced and adaptable protection mechanism against emerging threats is provided by combining the selection of K-Best features with ensemble modeling, which significantly strengthens the security of IoT [53].

4. Result and Discussion

We use the F1 score, recall, accuracy, and precision in our thorough evaluation of machine learning models for network intrusion detection. One way to evaluate a model's performance is by looking at its accuracy in predicting actual results. An essential aspect of intrusion detection is avoiding false positives, and precision measures how reliable negative predictions are. The true positive rate, also known as recall, is a metric for evaluating how well a model can detect positive examples, in this case real network intrusions. To ensure a strong intrusion detection system with minimal missed detection or false negatives, it is crucial to achieve high recall values.

An aggregate measure of the efficacy of the model, F1 is a harmonic average of recall and precision. A higher F1 shows better performance when accuracy and completeness are considered. Our network intrusion detection technologies that rely on machine learning are thoroughly evaluated using these carefully chosen KPIs. To keep things fair, we highlight the model's capacity to correctly detect and minimize possible dangers while concentrating on recall and precision. In order to provide a nuanced understanding of the model's performance, our review comprises a thorough analysis of these measures. Our aim is to provide an open and thorough assessment of the ensemble model's performance in the complex field of network intrusion recognition by taking all of these elements into account. Fig. 4 and Table I show the results of the performance evaluation of the suggested models using Accuracy, Recall, Precision, and F1-Score.

Table I. Performance evaluation

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
RF	98.3	98.2	98.19	98.18
NB	80.5	87.45	81.35	81.45
SVM	98.1	95.2	98.10	98.09
KNN	99.9	99.9	99.91	99.9
Stack Classifier	99.99	99.98	99.99	99.98

Fig. 4 prominently displays the results of trained heterogeneous stack classifier model on the ToN-IoT dataset. In this evaluation, we will examine these outcomes. The suggested stack classifier achieved an astonishing 99.99% accuracy rate, proving its remarkable competence in correctly classifying instances of network traffic as normal or intrusive. Complementing this outstanding accuracy was a precision of 99.98%, ensuring continued high accuracy in intrusion detection. In addition, the model demonstrated a remarkable recall rate of 99.99%, indicating its ability to correctly detect actual incursions while maintaining an exceptionally low level of false negatives. The stack classifier had exceptional overall performance with an F1 score of 99.98%, which is a measure of precision and recall combined.

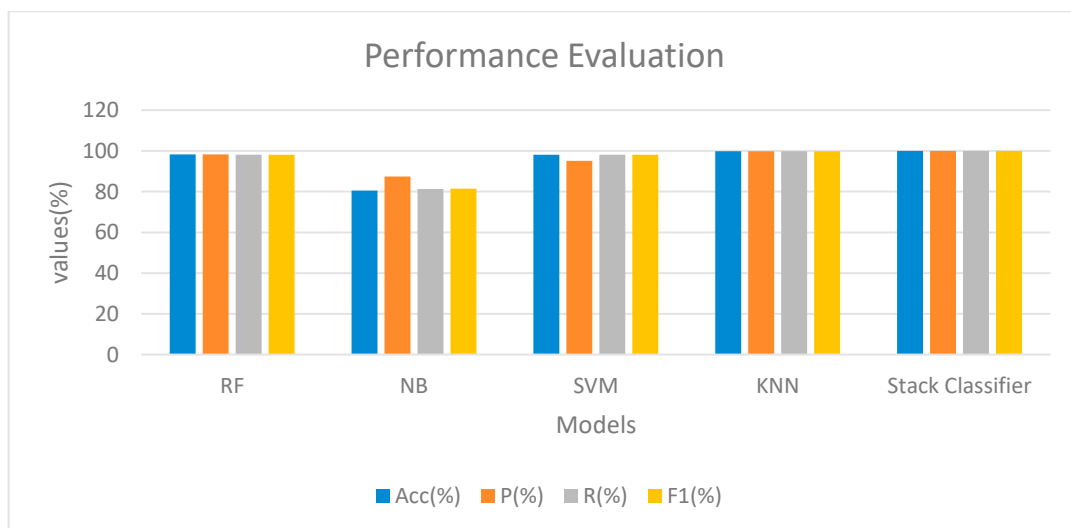


Fig. 4. Performance evolution of the proposed models

Table II shows the results of our accuracy comparison with DeepAK-IoT [54], LSTM [55], FFNN [56] and RaaNN [57], as well as with our suggested stack classifier model.

Table II Comparative analysis	
Model	Accuracy (%)
DeepAK-IoT	90.59
LSTM	99.51
FFNN	99.23
RaaNN	96.42
Stack Classifier	99.99

Fig. 5. shows the graphical representation of the precision comparisons. On the contrary, our ensemble method achieved astonishing accuracy ratings ranging from 80.46% to 99.99% by using a variety of models. Among the competing models, our stack classifier model had the best overall performance with a 99.99% accuracy rate.

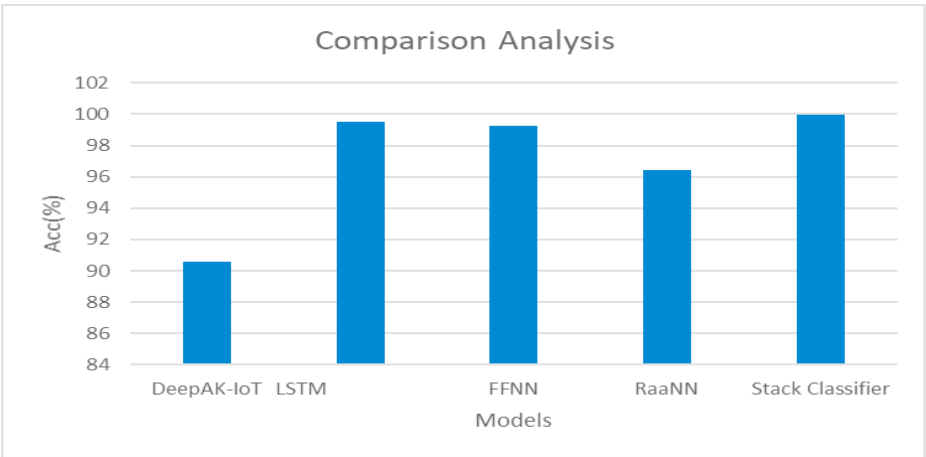


Fig. 5. Comparison analysis

Finally, when looking at important metrics like accuracy, precision, recall, and F1 score, the comparative results shows that the stack classifier consistently outperforms individual models like KNN, RF, SVM, and NB. The stack classifier is closely followed by KNN, RF and SVM both show strong results, and NB is behind with rather poor accuracy. We appreciate the importance of considering several metrics and appropriate aspects when evaluating the effectiveness of the model, and we also recognize the vital importance of accuracy. Researchers, practitioners, and decision-makers can all benefit from this exhaustive analysis, which details needs, performance objectives, and trade-off preferences to help them make well-informed decisions.

The stack classifier is attractive for real-world deployment because of its technical competence and its practical versatility. We prove its continuous dominance over individual models and illuminate its broader implications as our investigation progresses. The Accuracy, Precision, Recall, and F1-score balance demonstrated by the ensemble approach's detailed performance evaluation demonstrates its robustness in eliminating false positives and false negatives. The stack classifier is a priceless tool for intrusion detection in the complex world of IoT contexts because it prioritizes both technical perfection and practical efficacy. Providing a complete and efficient answer to the problem of Internet of Things (IoT) network security, the thorough study goes beyond technical measurements. Researchers, practitioners, and decision-makers seeking enhanced security in complex IoT environments can rely on the Stack Classifier's consistent high performance across precision, recall, and F1-score, along with exceptional accuracy. With the help of the in-depth analysis presented here, you can choose a model that fits your unique needs in terms of security and performance.

5. Conclusion and future work

This research has introduced a robust stack classification model that successfully tackles IoT network's essential intrusion detection problem. We have optimized key classification metrics to identify intrusions in IoT data with our model, which is systematically built to combine ensemble modelling and the K-Best feature selection with a Stack Ensemble Classifier technique. We have successfully captured significant information from these variables and harnessed the capabilities of each model by strategically selecting the top 15 critical features and adeptly integrating multiple classic machine learning models into an ensemble. Our comprehensive experimental tests, which compared our ensemble model to individual models and used the "Ton IoT dataset" frequently showed that it performed exceptionally well. By routinely surpassing competitors in Accuracy, Precision, Recall, and F1-score, the ensemble model solidified its position as a strong response to the problem of reducing security risks in IoT networks.

Our future research objectives will be improving the flexibility of our ensemble model for various IoT networks and devices, investigating scaling possibilities, and guaranteeing seamless integration into diverse infrastructures. This focused effort aims to make the model as applicable as possible so that it can tackle the specific problems that arise in different kinds of IoT scenarios.

Availability of Data and Materials

The data and supporting information are available in the article.

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Conflict of Interest

The authors declare no conflict of interest.

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Authors' Contribution

It is hereby acknowledged that all authors have accepted responsibility for the manuscript's content and consented to its submission. They have meticulously reviewed all the results and unanimously approved the final version of the manuscript.

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